
Algorithm 1: TOPNet Network Construction

Input: $\theta = (\theta_1, \dots, \theta_{10}) \in [-0.5, 0.5]^{10}$, $n \times n$ tiling, tiling_mode, dimension, target metric

Output: $G(V, E)$ (Eulerian network graph), P (continuous path)

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1 Step 1: Eulerian Base-Unit Generation;
2  $U_0 \leftarrow$  Initialize square grid template;
3 foreach edge  $e \in \{top, bottom, left, right\}$  do
4   | Define 5 control points:  $c_e = [c_1, \dots, c_5]$ ;
5 end
6 Apply displacement  $\Delta(\theta)$  to one edge, replicate with rotational symmetry;
7 Verify  $d(v) \equiv 0 \pmod{2}$  for all nodes  $v \in U_0$ ;
8  $U_{\text{def}} \leftarrow$  Deformed unit;
9 Step 2: Regular Tiling;
10 if tiling_mode = uniform then
11   |  $G \leftarrow$  Replicate  $U_{\text{def}}$  in  $n \times n$  matrix;
12 else
13   | Generate  $U'_{\text{def}}$  with  $\theta'$ , arrange in checkerboard pattern;
14 end
15 Merge boundaries, split overlapping edges, verify  $d(v) \equiv 0 \pmod{2}$ ;
16 Step 3: Anchor-Point Regulation;
17  $M \leftarrow$  Project fibers onto 2D matrix,  $\rho_{\text{current}} \leftarrow N_{\text{anchor}}/A_{\text{domain}}$ ;
18 while  $|\rho_{\text{current}} - \rho_{\text{tar}}| > \text{tolerance}$  do
19   | Optimize  $\theta$ , rebuild  $U_{\text{def}}$ , update  $G$ , recompute  $\rho_{\text{current}}$ ;
20 end
21 Step 4: Boundary Fitting;
22 if dimension = 2D then
23   | Map perimeter nodes to Contour, apply Laplacian smoothing;
24 else
25   | // 3D: QuadriFlow-based surface mapping
26   | Apply QuadriFlow on target surface  $S_{\text{target}}$  to get quad mesh  $Q$ ;
27   | Project boundary nodes onto  $S_{\text{target}}$ , optimize interior via energy minimization;
28 end
29 Step 5: Canonical Grid Mapping (CGM);
30  $G_{\text{canonical}} \leftarrow$  Create  $3 \times 3$  reference grid;
31 Find Eulerian path, map node sequence to deformed  $G$  coordinates;
32  $P \leftarrow$  Connect mapped coordinates into continuous path;
33 return  $G(V, E), P$ ;
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Algorithm 2: RL-based Inverse Design with SAC

Input: Target $\{\text{LC150}_{\text{tar}}, \text{mass}_{\text{tar}}\}$, mode $\in \{\text{Loss-Weight}, \text{Fixed-Weight}\}$
Output: θ^* (optimal encoding)

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1 Initialize: Actor  $\pi_\phi$ , Critics  $Q_{\psi_1}, Q_{\psi_2}$ , Targets  $\bar{Q}$ , Buffer  $\mathcal{D}$ , Temp  $\alpha$ ;  
2 for  $\text{episode} = 1$  to  $N_{\text{episode}}$  do  
3    $\theta_0 \sim \text{Uniform}[-0.5, 0.5]^{10}$ ,  $s_0 \leftarrow \text{Encode}(\theta_0, \text{features}, \text{state})$ ;  
4   for  $\text{step } t = 1$  to  $T$  do  
5     Action Selection:  $\mu, \log \sigma \leftarrow \pi_\phi(s_t)$ ,  $\epsilon \sim \mathcal{N}(0, I)$ ;  
6      $\theta_t \leftarrow \tanh(\mu + \sigma \odot \epsilon)$ ;  
7     if  $\|\theta_t\|_2 > 0.5$  then  
8        $\theta_t \leftarrow 0.5 \cdot \theta_t / \|\theta_t\|_2$ ;  
9     end  
10    Environment:  $G, P \leftarrow \text{TOPNet}(\theta_t)$ , verify connectivity;  
11    Extract features  $\{S_1, \dots, S_{12}\}$ , LC150, mass  $\leftarrow \text{GNN}(G)$ ;  
12    Reward::  
13    if  $\text{mode} = \text{Loss-Weight}$  then  
14       $r \leftarrow (\text{LC150}/\text{LC150}_{\text{tar}}) - \lambda_w \cdot (\text{mass}/\text{mass}_{\text{tar}})$ ;  
15      if  $\text{LC150} \geq \text{target}$  and  $\text{mass} \leq \text{target}$  then  
16         $r \leftarrow r + h_b$ ;  
17      end  
18    else  
19       $\text{gap} \leftarrow |\text{mass} - 6.5|$ ;  
20      if  $\text{gap} \leq 0.05$  then  
21         $r \leftarrow h_b + \text{LC150}$ ;  
22      else  
23         $r \leftarrow \text{LC150} - \lambda_{\text{penalty}} \cdot (\text{gap} - 0.05)^2$ ;  
24      end  
25    end  
26     $s_{t+1} \leftarrow \text{Encode}(\theta_t, \text{features}, \{\text{LC150}, \text{mass}\})$ ;  
27    Store  $(s_t, \theta_t, r, s_{t+1})$  in  $\mathcal{D}$ ;  
28    SAC Update::  
29    if  $|\mathcal{D}| \geq \text{batch\_size}$  then  
30      Sample batch  $\mathcal{B}$ , compute target:  $y \leftarrow r + \gamma(\min \bar{Q}_{\psi_i}(s', \theta') - \alpha \log \pi_\phi(\theta' | s'))$ ;  
31      Update critics:  $\min L(\psi_i) = \mathbb{E}[(Q_{\psi_i}(s, \theta) - y)^2]$ ;  
32      Update actor:  $\max J(\phi) = \mathbb{E}[\min Q_{\psi_i}(s, \theta) - \alpha \log \pi_\phi(\theta | s)]$ ;  
33      Soft-update:  $\bar{\psi}_i \leftarrow \tau \psi_i + (1 - \tau) \bar{\psi}_i$ , update  $\alpha$ ;  
34    end  
35    if  $\text{Termination}(\text{LC150}, \text{mass}, \text{target})$  then  
36      break;  
37    end  
38  end  
39 end  
40 return  $\theta^* \leftarrow \arg \max_\theta \mathbb{E}[Q_\psi(s, \theta)]$ ;
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